Plot some time series of Speedy results

In [44]:
```python
import numpy as np
import xarray as xr
from matplotlib import pyplot as plt
import pandas as pd
plt.rcParams['figure.figsize'] = (12, 6)
%config InlineBackend.figure_format = 'retina'
%xmode Minimal

import warnings
warnings.filterwarnings('ignore')
```

```
Exception reporting mode: Minimal
```

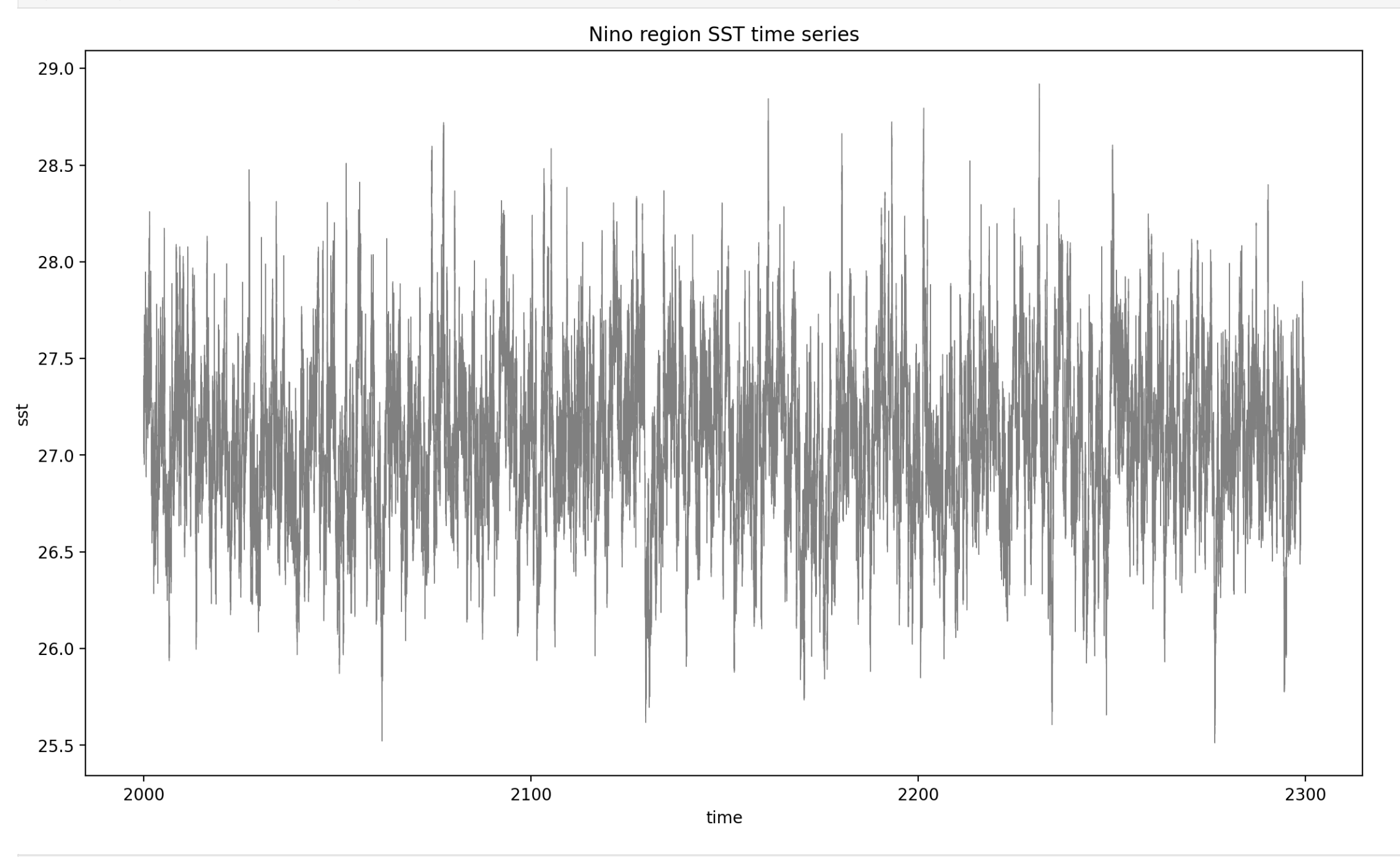
In [47]:
```python
run='Climate_Ocean_Land'
runn=run+'_0003'
data_file = '/Users/vidale/SpeedyWeather/run_'+runn+'/output.nc'
#data_file = '/Users/vidale/SpeedyWeather/output_run'+runn+'.nc'

#open data using x_array
data = xr.open_dataset(data_file)

#slice the data from
data = data.sel(time=slice('2000','2299'))
data
```

Out[47]:
```
xarray.Dataset

▶ Dimensions:      (time: 876001, lon: 64, lat: 32, layer: 8, soil_layer: 2)

▼ Coordinates:
   time          (time)          object  2000-01-01 00:00:00 ... 2299-10-...
   lon           (lon)          float64  0.0 5.625 11.25 ... 348.8 354.4
   lat           (lat)          float64  85.76 80.27 74.74 ... -80.27 -85.76
   layer         (layer)        float32  0.0625 0.1875 ... 0.8125 0.9375
   soil_layer    (soil_layer)     int64  1 2

▶ Data variables:
 (28)

▶ Indexes:  (5)

▶ Attributes:  (0)
```

In [3]:
```python
#nino_region
nino_region = data.sel(lon=slice(190, 240), lat=slice(6,-6))
```

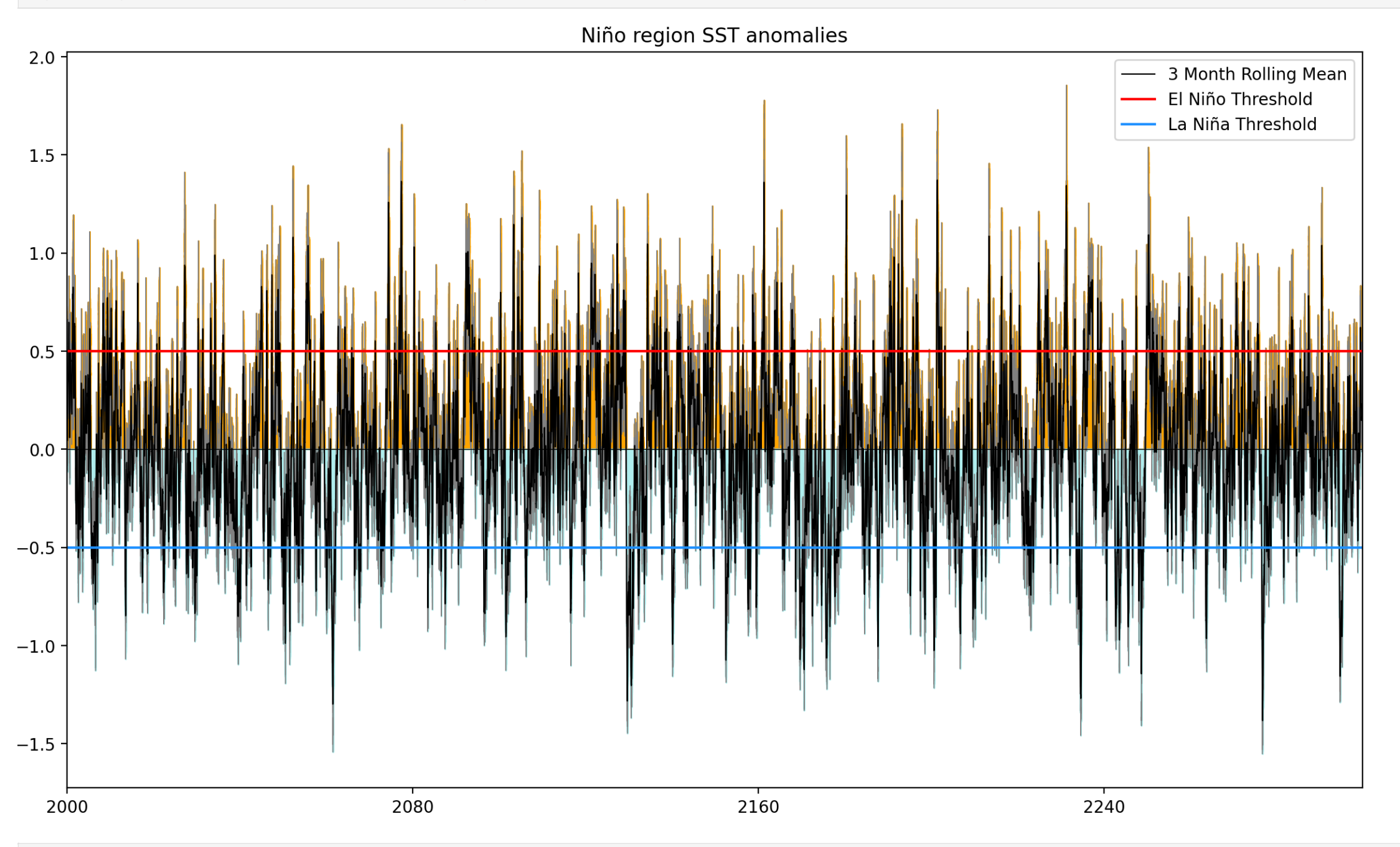
In [36]:
```python
#the mean over the entire area (spatially)
nino_region_mean = nino_region.mean(dim=('lat','lon'))

#the spatial and temporal mean for the entire area
nino_region_mean_climate = nino_region_mean.sel(time=slice('2000-01-01', '2029-12-31')).mean()

#build the anomoly dataset
nino_anom = (
    nino_region_mean['sst'].sel(time=slice('2000-01-01', '2299-10-01')) -
    nino_region_mean_climate['sst']
)
```

In [37]:
```python
#plot the means
fig=plt.figure(figsize=(14,8))
nino_region_mean.sst.plot(linewidth=0.5, color='grey')
plt.title("Nino region SST time series")
#show plot
plt.show()
fig.savefig("NinoSST_run_"+runn+".png")
```

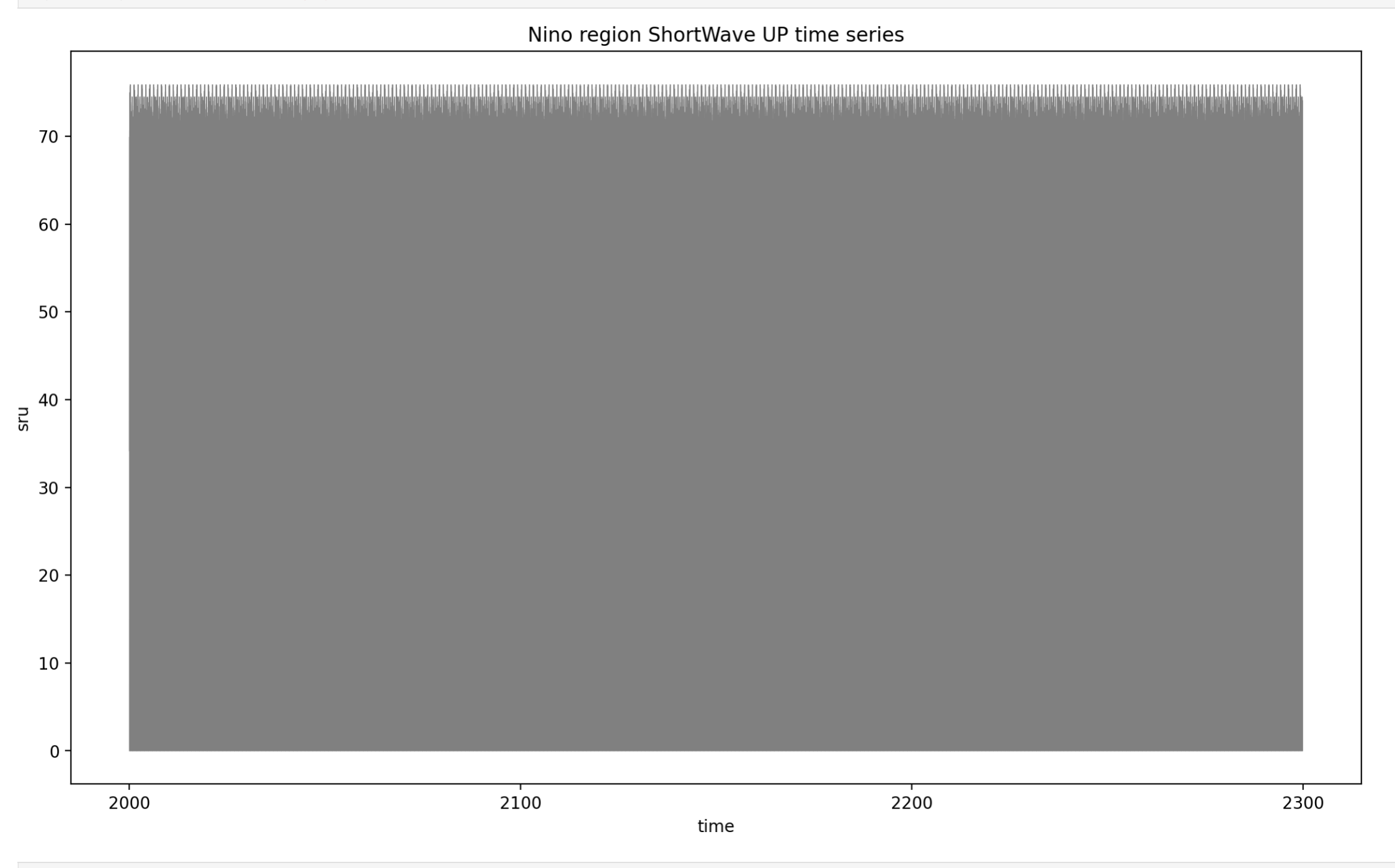


In [38]:
```python
#create a rolling 3 month mean dataset
#rolling_mean = nino_anom.rolling(time=3, center=True).mean()
rolling_mean = nino_anom.rolling(time=360, center=True).mean()
#define mask for positive and negative anomoly values
positive = nino_anom.where(nino_anom > 0)
negative = nino_anom.where(nino_anom < 0)
```

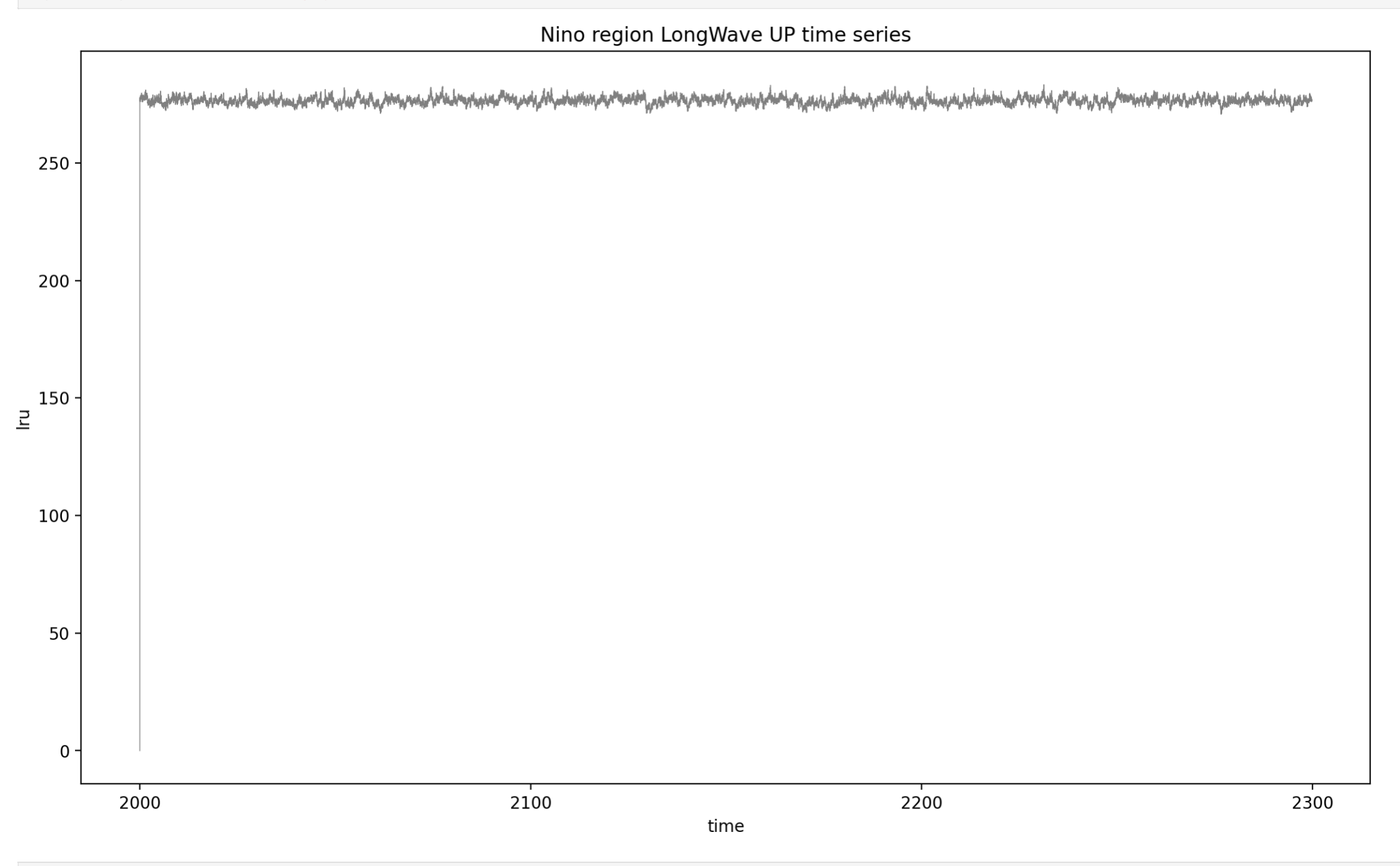
In [48]:
```python
fig = plt.figure(figsize=(14, 8))

# Assume you already have these:
# - nino_anom: DataArray of anomalies
# - rolling_mean: 3-month rolling mean of nino_anom

# Compute positive/negative parts for fill
positive = nino_anom.where(nino_anom > 0, 0)
negative = nino_anom.where(nino_anom < 0, 0)

# Plot anomalies as a thin gray line
plt.plot(nino_anom['time'].values, nino_anom.values, color='gray', linewidth=0.5)

# Fill between 0 and positive/negative anomalies
plt.fill_between(nino_anom['time'].values, 0, positive.values, color='orange')
plt.fill_between(nino_anom['time'].values, 0, negative.values, color='paleturquoise')

# Plot rolling mean
plt.plot(rolling_mean['time'].values, rolling_mean.values, color='k', linewidth=0.8, label='3 Month Rolling Mean')

# Reference lines
plt.axhline(0, linewidth=0.5, color='black')
plt.axhline(0.5, color='red', label='El Niño Threshold')
plt.axhline(-0.5, color='dodgerblue', label='La Niña Threshold')

# Set limits and labels
plt.xlim(nino_anom['time'].values[0], nino_anom['time'].values[-1])
plt.legend(bbox_to_anchor=(1, 1))
plt.title("Niño region SST anomalies")

plt.show()
fig.savefig("NinoSSTAnomalies_run_" + runn + ".png")
```



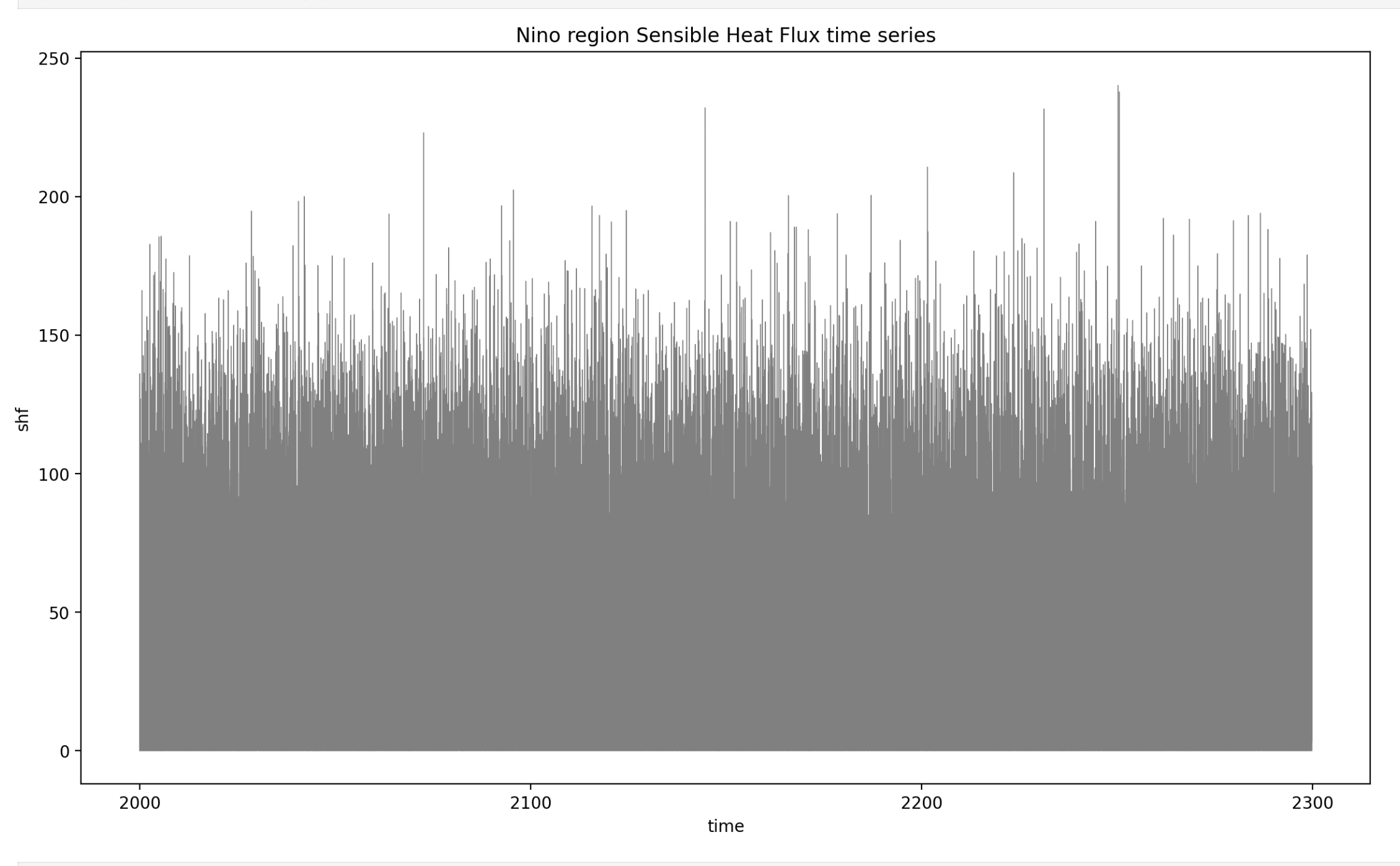
In [8]:
```python
#plot the means
fig=plt.figure(figsize=(14,8))
nino_region_mean.sru.plot(linewidth=0.5, color='grey')
plt.title("Nino region ShortWave UP time series")
#show plot
plt.show()
fig.savefig("SRU_run_"+runn+".png")
```



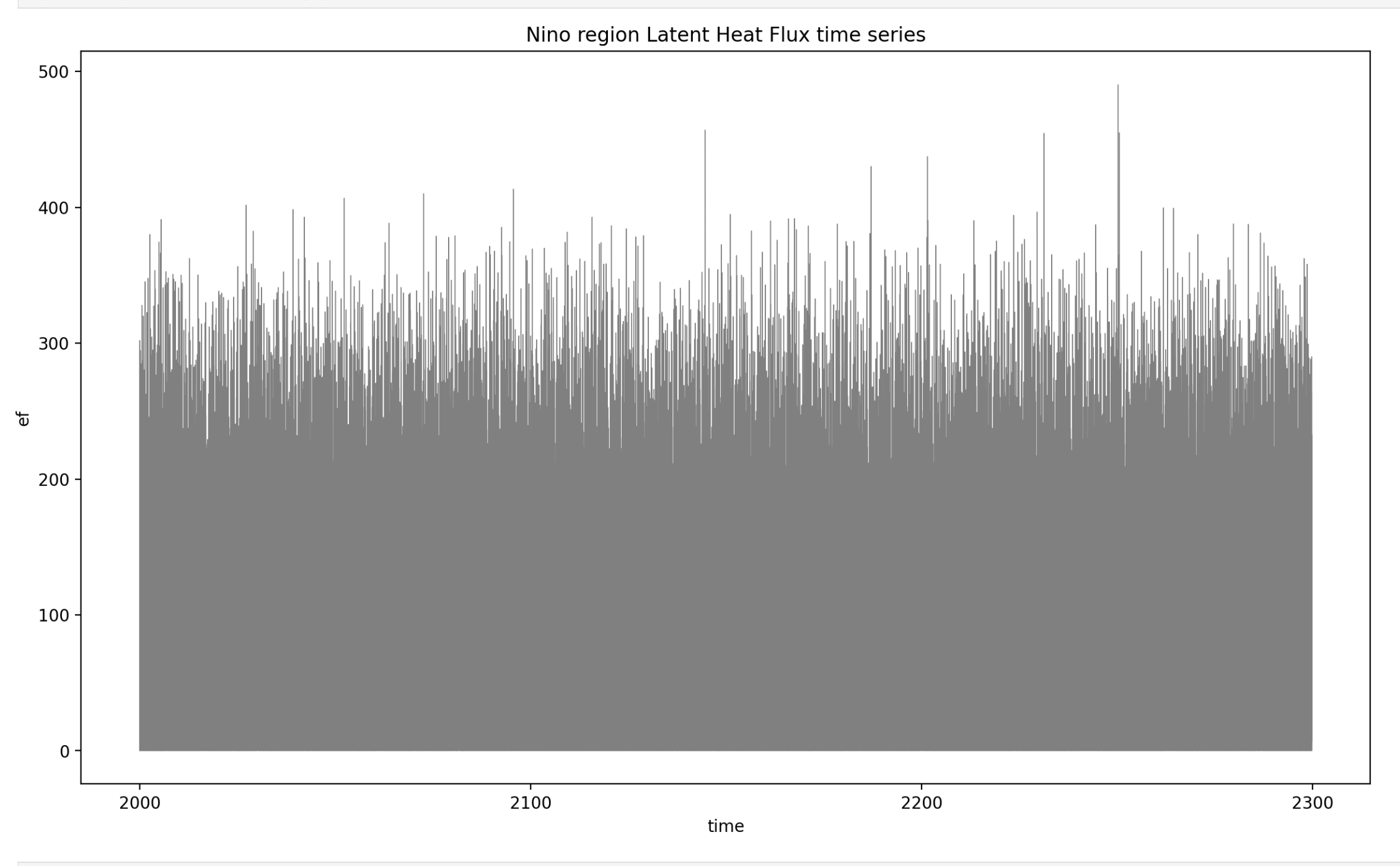
In [9]:
```python
#plot the means
fig=plt.figure(figsize=(14,8))
nino_region_mean.lru.plot(linewidth=0.5, color='grey')
plt.title("Nino region LongWave UP time series")
#show plot
plt.show()
fig.savefig("LRU_run_"+runn+".png")
```



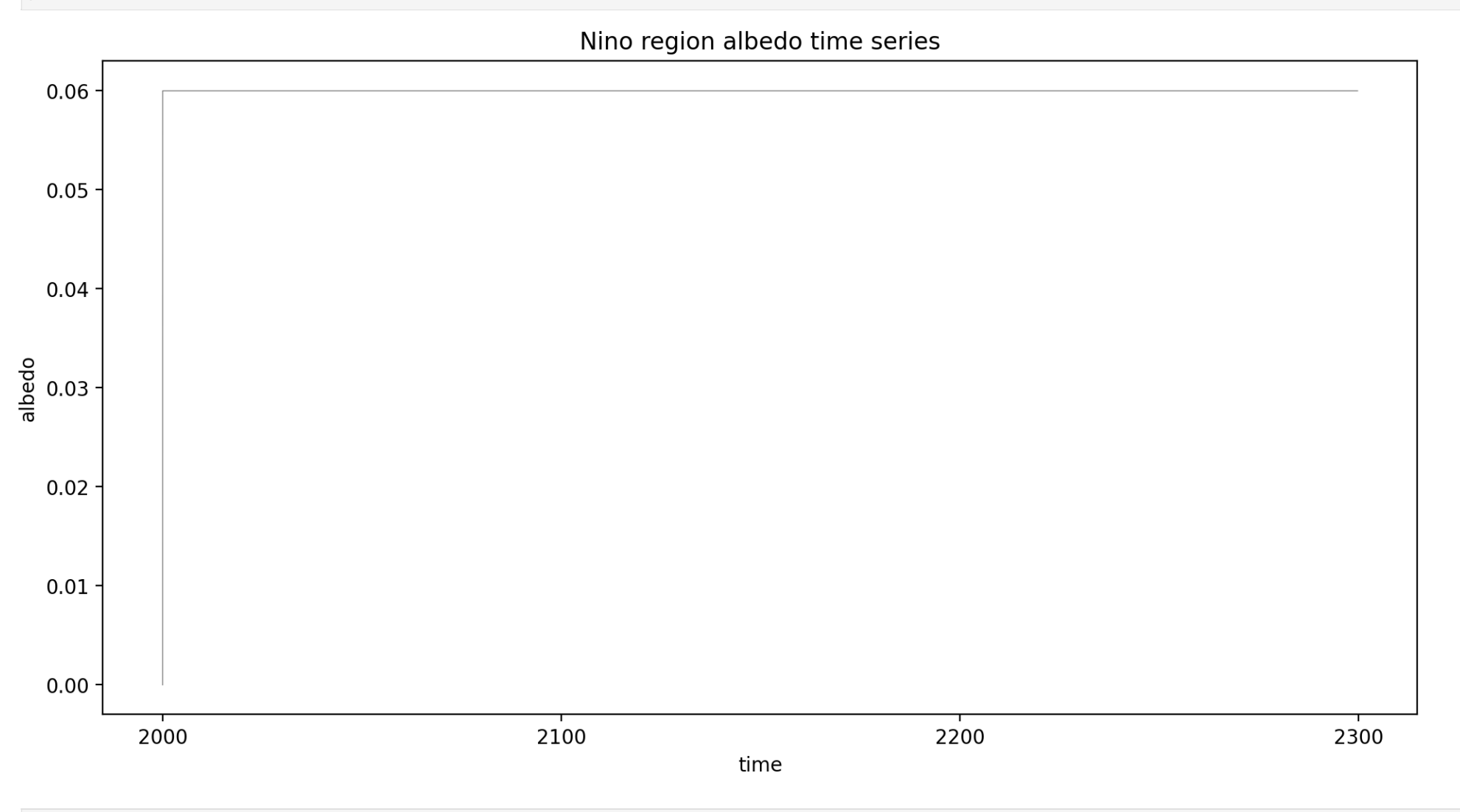
In [10]:
```python
#plot the means
fig=plt.figure(figsize=(14,8))
nino_region_mean.shf.plot(linewidth=0.5, color='grey')
plt.title("Nino region Sensible Heat Flux time series")
#show plot
plt.show()
fig.savefig("SHF_run_"+runn+".png")
```



In [11]:
```python
#plot the means
fig=plt.figure(figsize=(14,8))
lhf=2.5E6*nino_region_mean.ef
lhf.plot(linewidth=0.5, color='grey')
plt.title("Nino region Latent Heat Flux time series")
#show plot
plt.show()
fig.savefig("LHF_run_"+runn+".png")
```



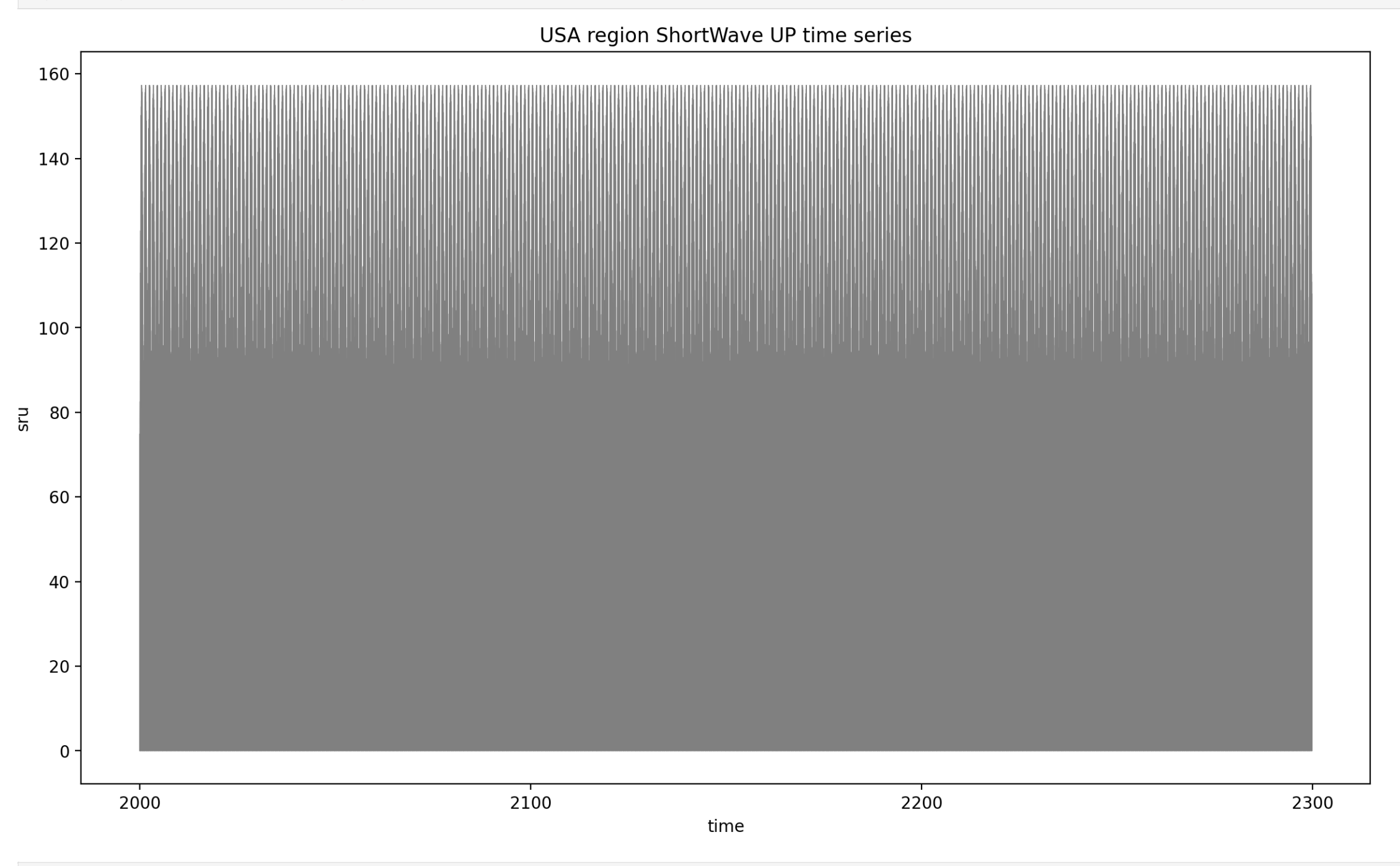
In [12]:
```python
#plot the means
nino_region_mean.albedo.plot(linewidth=0.5, color='grey')
plt.title("Nino region albedo time series")
#show plot
plt.show()
```



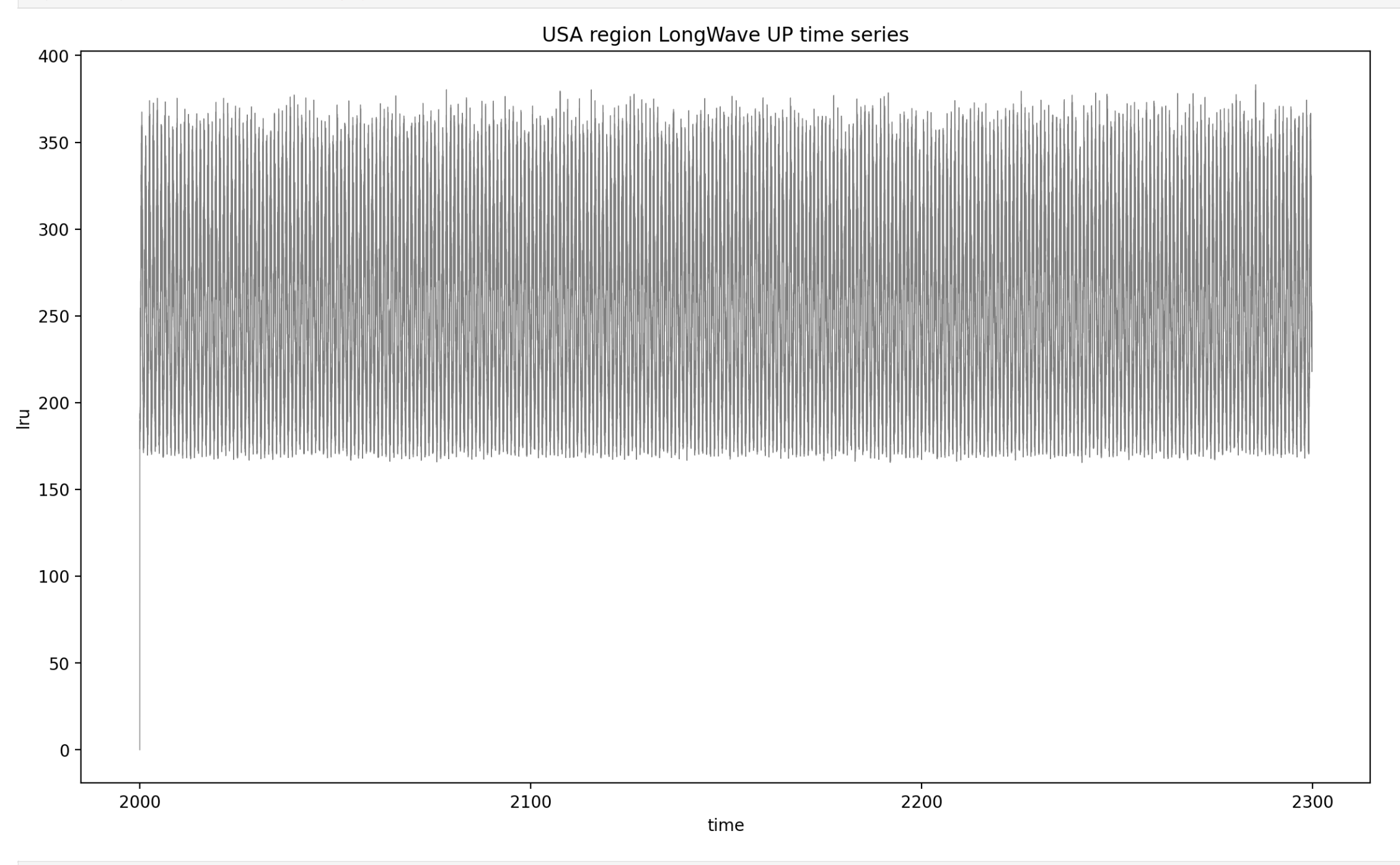
In [13]:
```python
USA_region = data.sel(lon=slice(236, 294), lat=slice(50, 25))
#USA_region
```

In [14]:
```python
#the mean over the entire area (spatially)
USA_region_mean = USA_region.mean(dim=('lat','lon'))

#the spatial and temporal mean for the entire area
USA_region_mean_climate = USA_region_mean.mean()

#build the anomoly dataset
#USA_anom = USA_month_mean - nino_region_mean
```

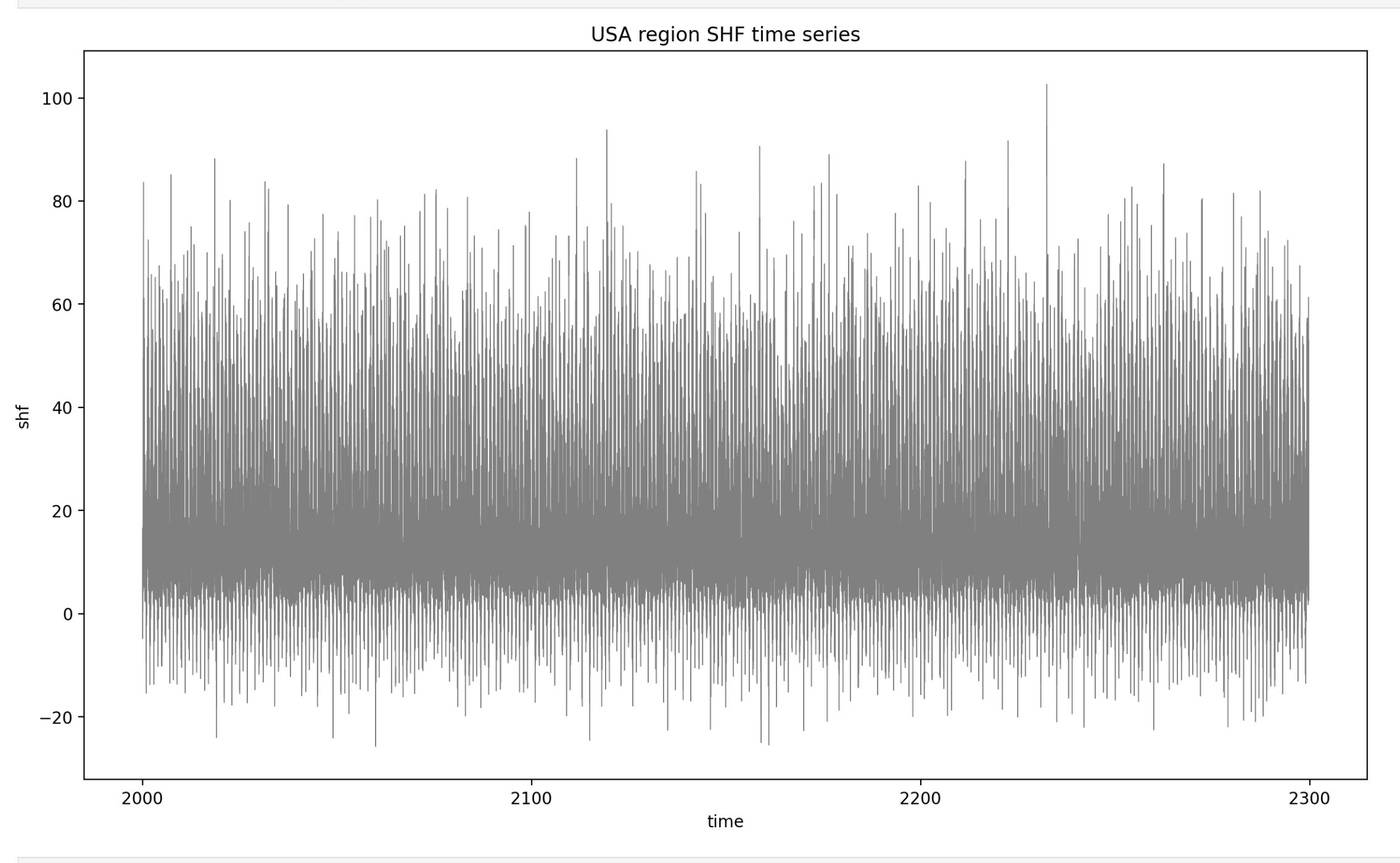
In [15]:
```python
#plot the means
fig=plt.figure(figsize=(14,8))
USA_region_mean.sru.plot(linewidth=0.5, color='grey')
plt.title("USA region ShortWave UP time series")
#show plot
plt.show()
fig.savefig("SRU_run_USA_"+runn+".png")
```



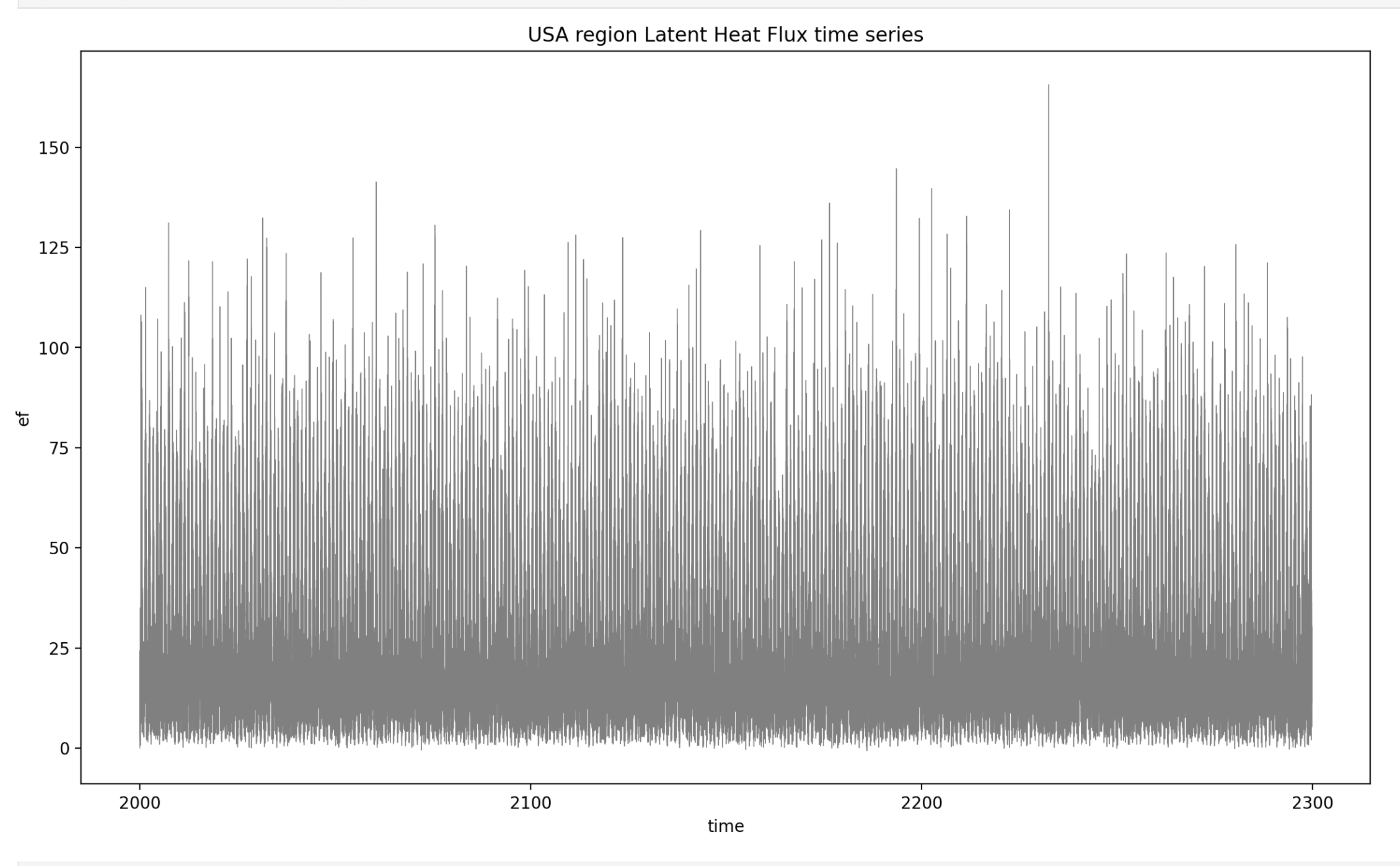
In [16]:
```python
#plot the means
fig=plt.figure(figsize=(14,8))
USA_region_mean.lru.plot(linewidth=0.5, color='grey')
plt.title("USA region LongWave UP time series")
#show plot
plt.show()
fig.savefig("LRU_run_USA_"+runn+".png")
```



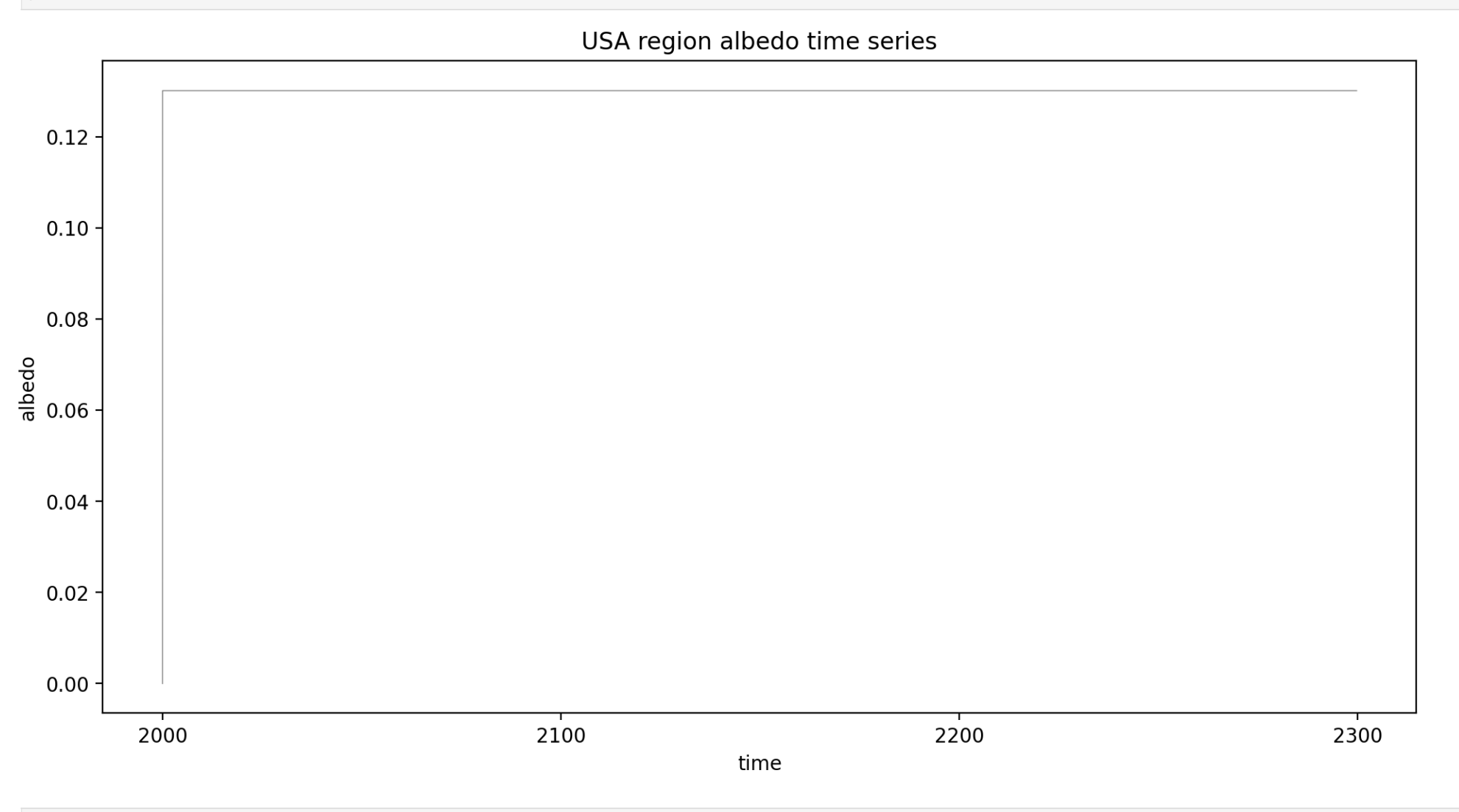
In [17]:
```python
#plot the means
fig=plt.figure(figsize=(14,8))
USA_region_mean.shf.plot(linewidth=0.5, color='grey')
plt.title("USA region SHF time series")
#show plot
plt.show()
fig.savefig("SHF_run_USA_"+runn+".png")
```



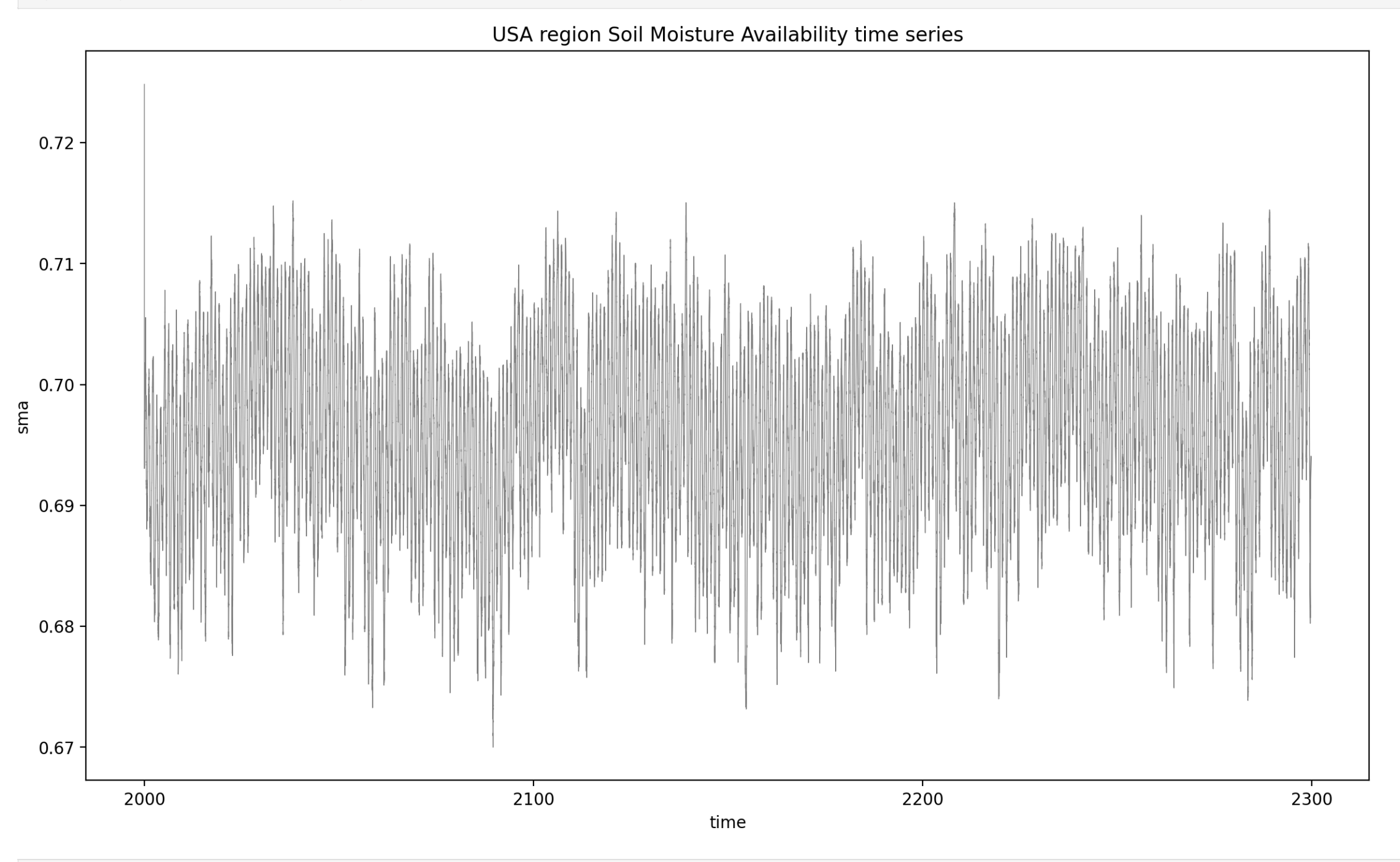
In [18]:
```python
#plot the means
fig=plt.figure(figsize=(14,8))
lhf=2.5E6*USA_region_mean.ef
lhf.plot(linewidth=0.5, color='grey')
plt.title("USA region Latent Heat Flux time series")
#show plot
plt.show()
fig.savefig("LHF_run_USA_"+runn+".png")
```



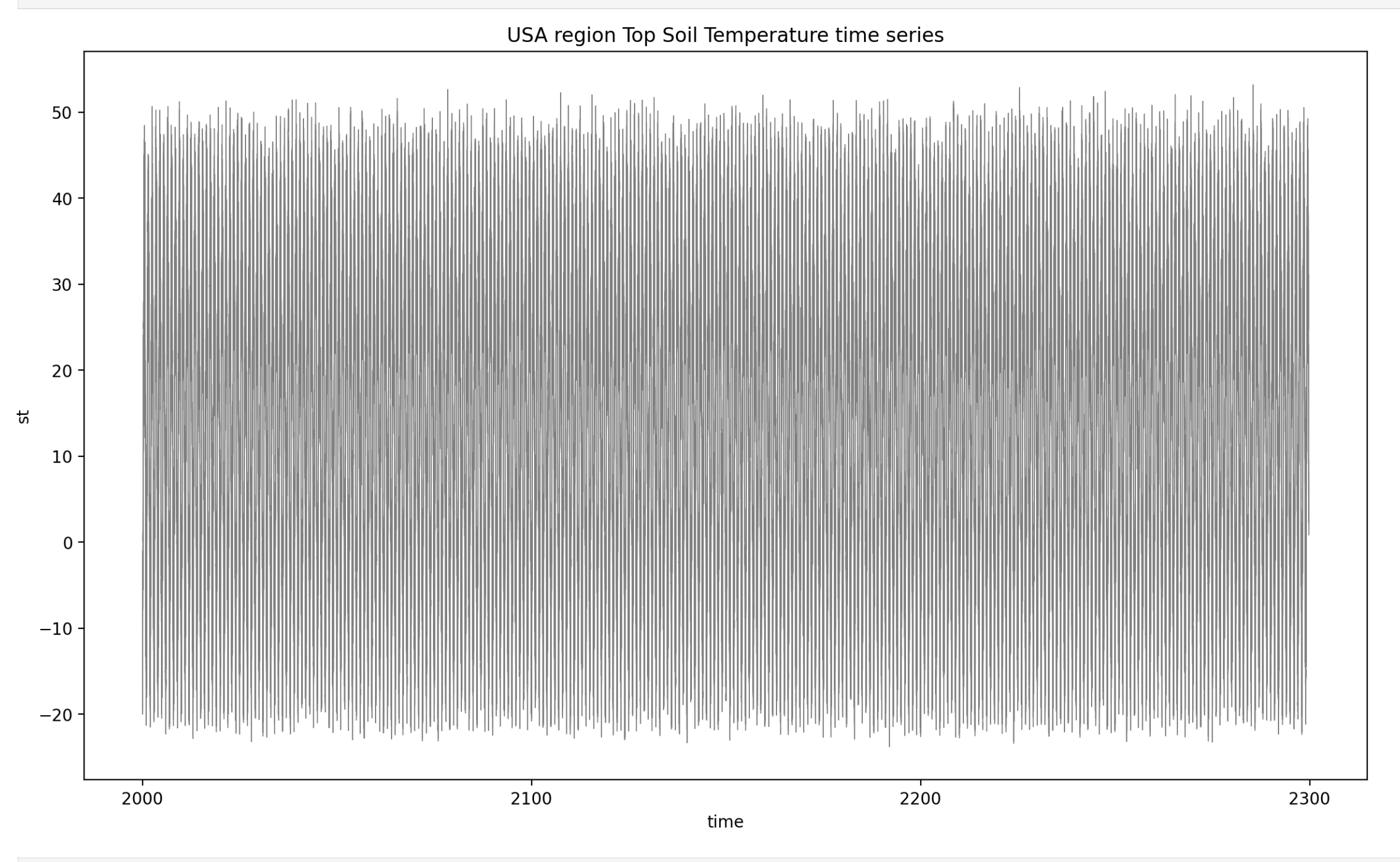
In [19]:
```python
#plot the means
USA_region_mean.albedo.plot(linewidth=0.5, color='grey')
plt.title("USA region albedo time series")
#show plot
plt.show()
```



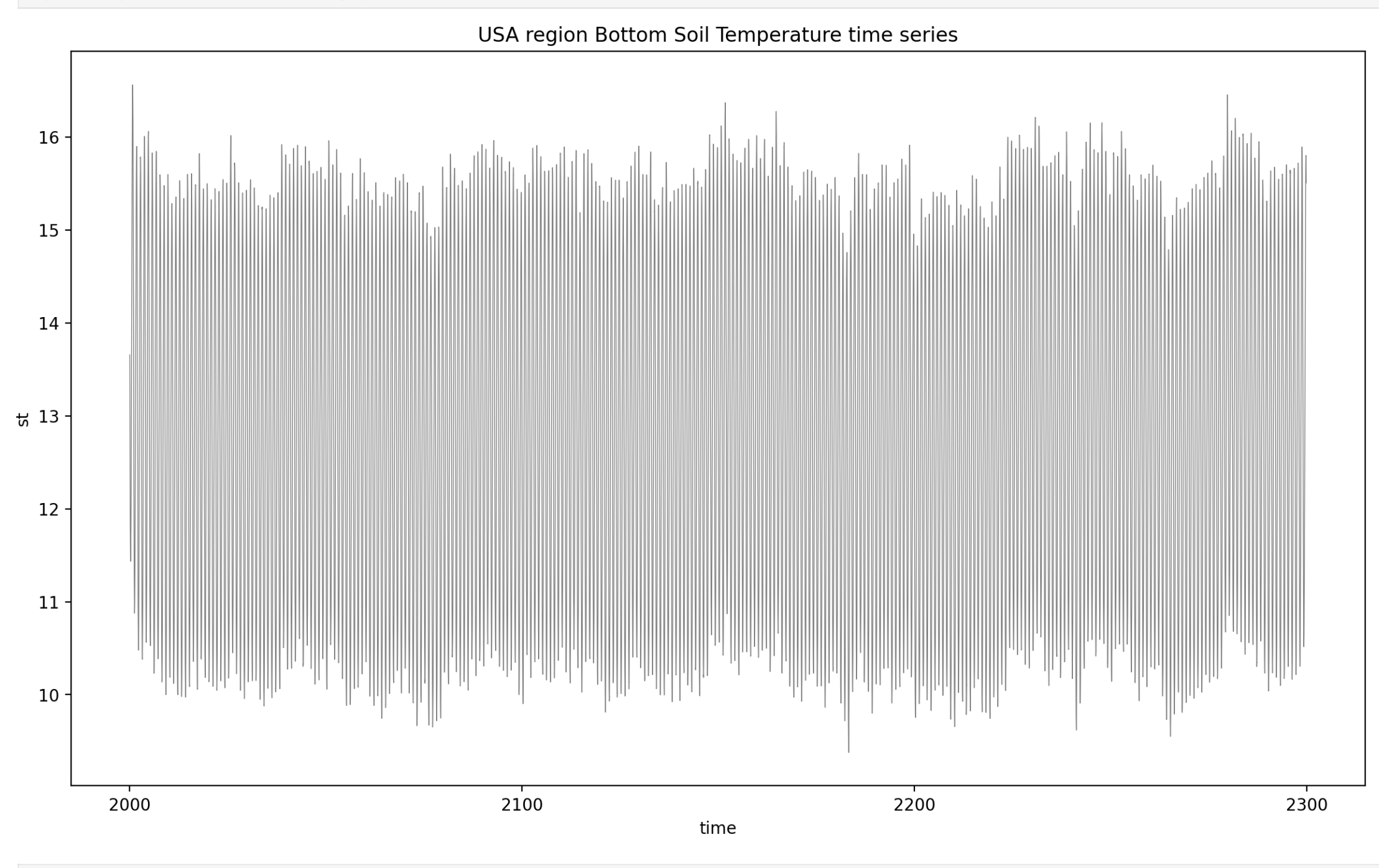
In [20]:
```python
fig=plt.figure(figsize=(14,8))
USA_region_mean.sma.plot(linewidth=0.5, color='grey')
plt.title("USA region Soil Moisture Availability time series")
#show plot
plt.show()
fig.savefig("SMA_run_USA_"+runn+".png")
```



In [21]:
```python
fig=plt.figure(figsize=(14,8))
st1=USA_region_mean.st[:,0]
st1.plot(linewidth=0.5, color='grey')
plt.title("USA region Top Soil Temperature time series")
#show plot
plt.show()
fig.savefig("ST1_run_USA_"+runn+".png")
```

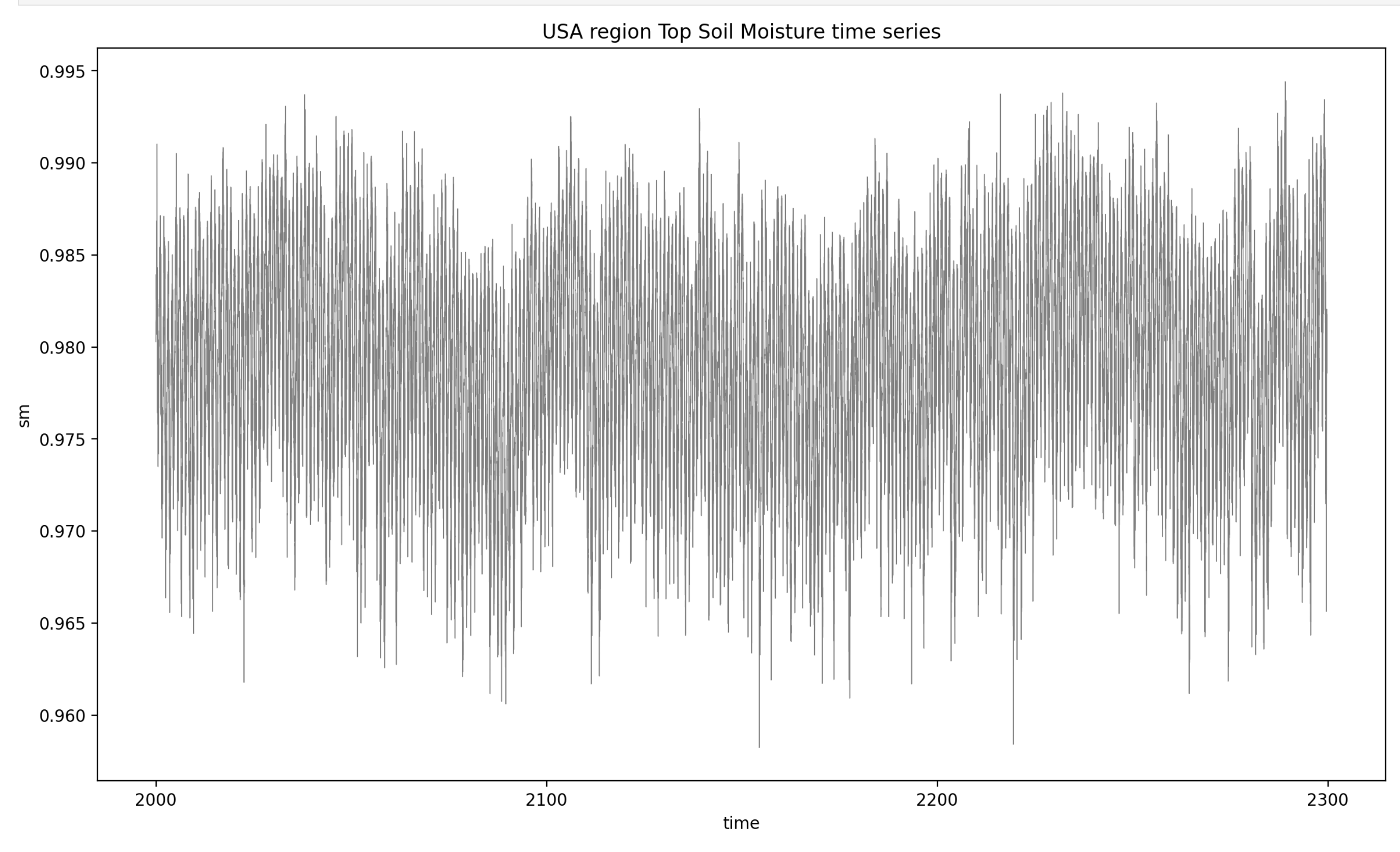


In [22]:
```python
fig=plt.figure(figsize=(14,8))
st2=USA_region_mean.st[:,1]
st2.plot(linewidth=0.5, color='grey')
plt.title("USA region Bottom Soil Temperature time series")
#show plot
plt.show()
fig.savefig("ST2_run_USA_"+runn+".png")
```



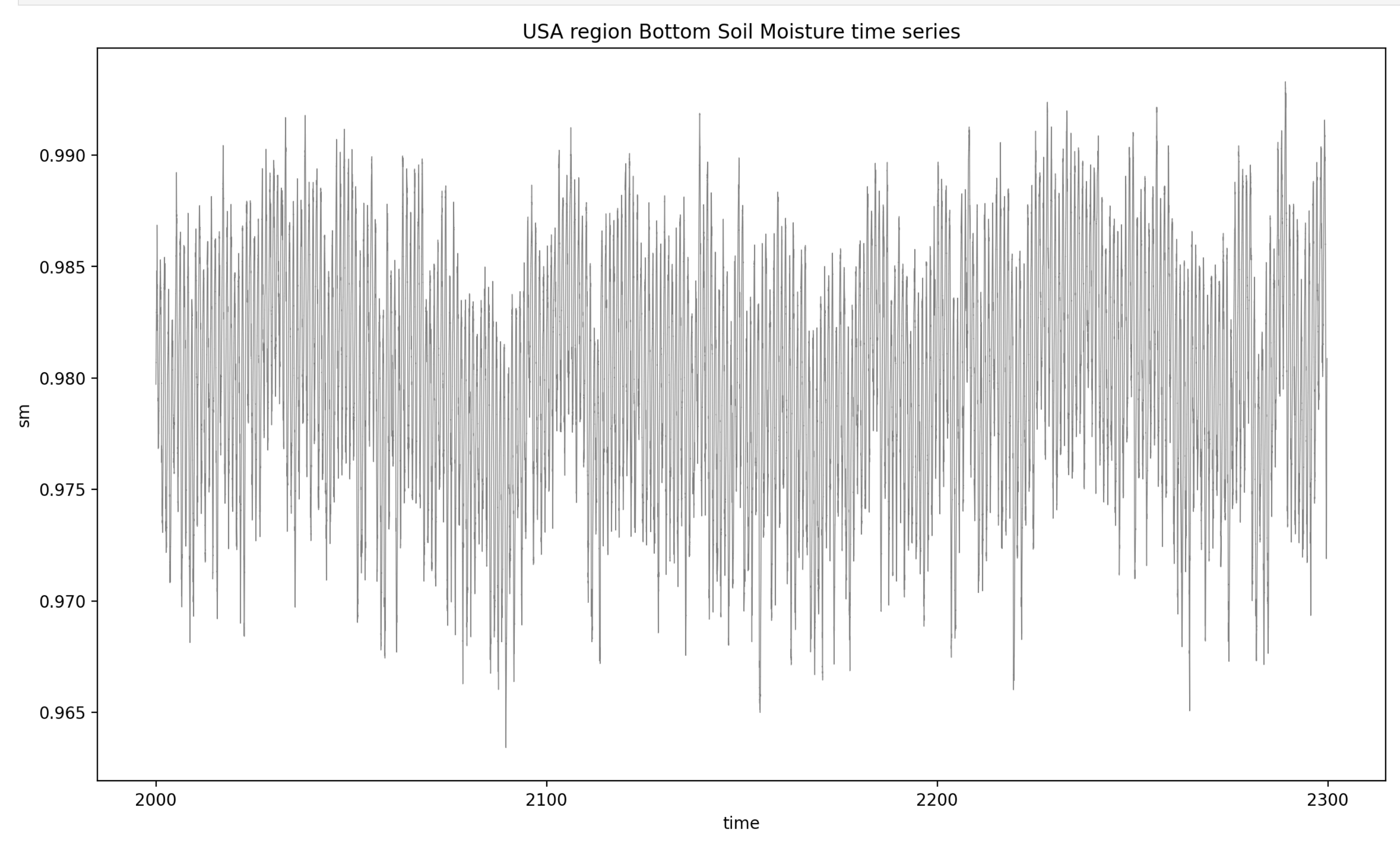
In [23]:
```python
fig=plt.figure(figsize=(14,8))
sm1=USA_region_mean.sm[:,0]
sm1.plot(linewidth=0.5, color='grey')
plt.title("USA region Top Soil Moisture time series")
#show plot
```

```
#show plot
plt.show()
fig.savefig("SM1_run_USA_"+runn+".png")
```



In [24]:

```
fig=plt.figure(figsize=(14,8))
sm2=USA_region_mean.sm[:,1]
sm2.plot(linewidth=0.5, color='grey')
plt.title("USA region Bottom Soil Moisture time series")
#show plot
plt.show()
fig.savefig("SM2_run_USA_"+runn+".png")
```



In [25]:

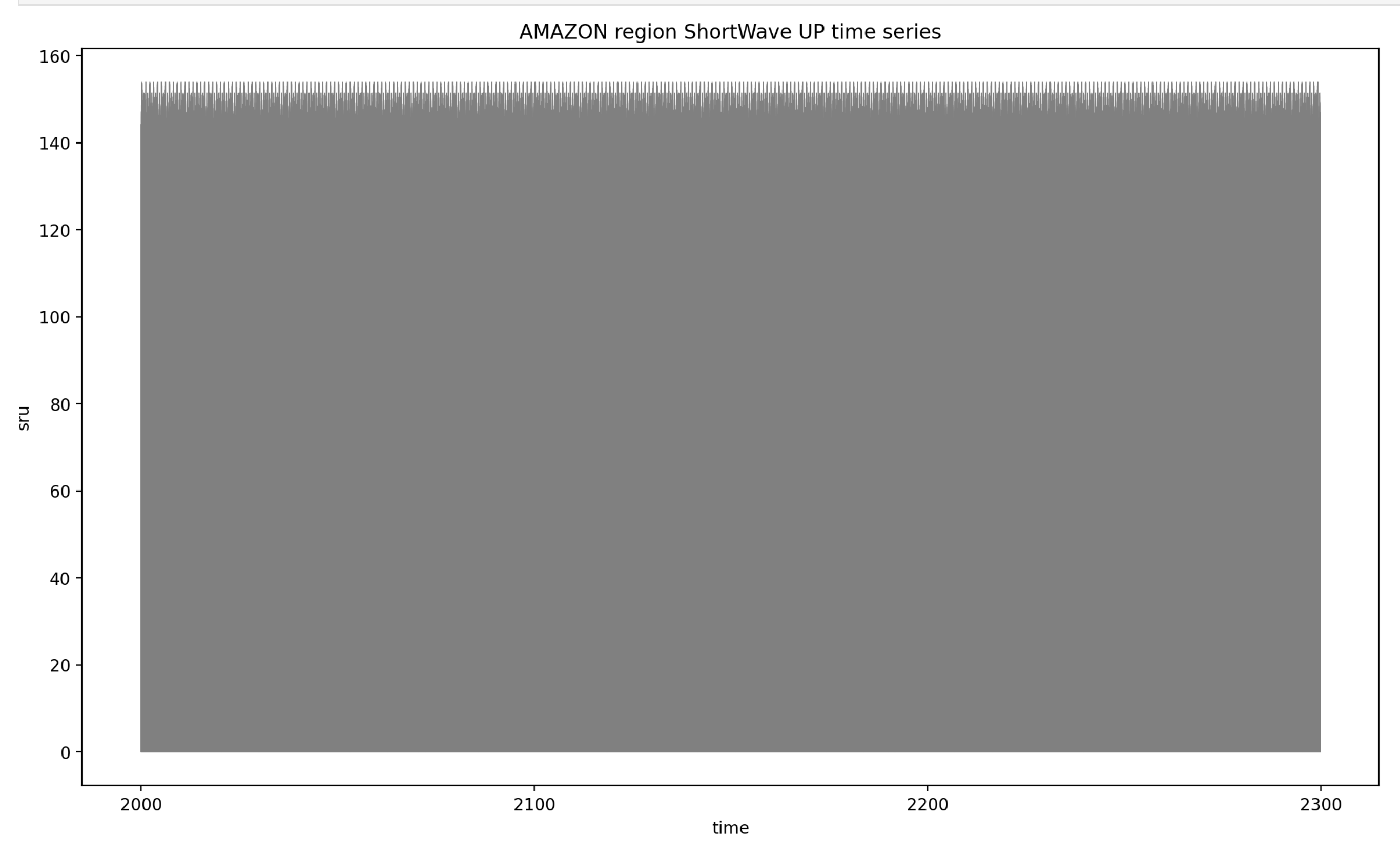
```
AMAZON_region = data.sel(lon=slice(285, 315), lat=slice(10, -10))
#AMAZON_region
```

In [26]:

```
#the mean over the entire area (spatially)
AMAZON_region_mean = AMAZON_region.mean(dim=('lat','lon'))

#the spatial and temporal mean for the entire area
AMAZON_region_mean_climate = AMAZON_region_mean.mean()
```

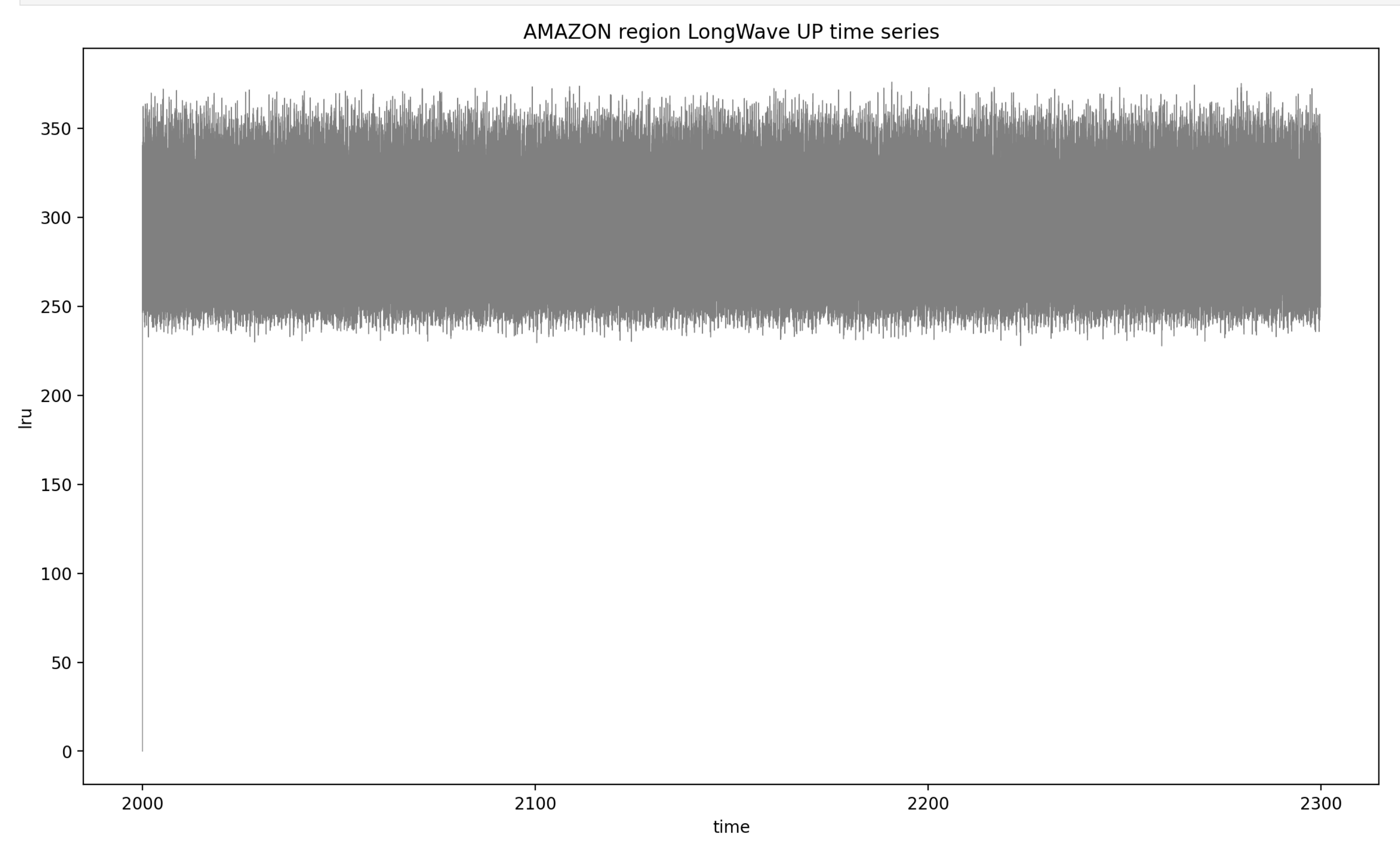
In [27]:

```
#plot the means
fig=plt.figure(figsize=(14,8))
AMAZON_region_mean.sru.plot(linewidth=0.5, color='grey')
plt.title("AMAZON region ShortWave UP time series")
#show plot
plt.show()
fig.savefig("SRU_run_AMAZON_"+runn+".png")
```



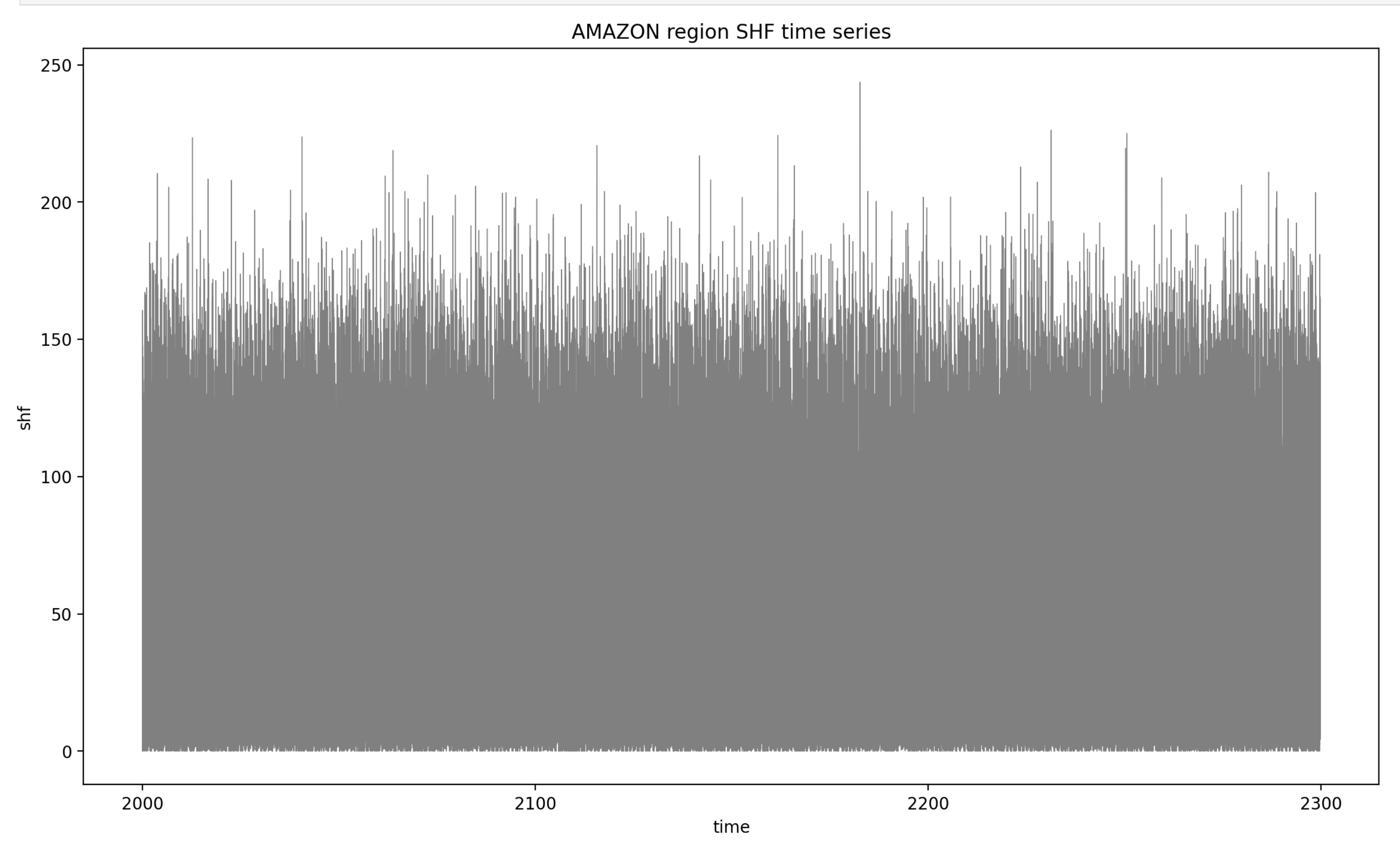
In [28]:

```
#plot the means
fig=plt.figure(figsize=(14,8))
AMAZON_region_mean.lru.plot(linewidth=0.5, color='grey')
plt.title("AMAZON region LongWave UP time series")
#show plot
plt.show()
fig.savefig("LRU_run_AMAZON_"+runn+".png")
```



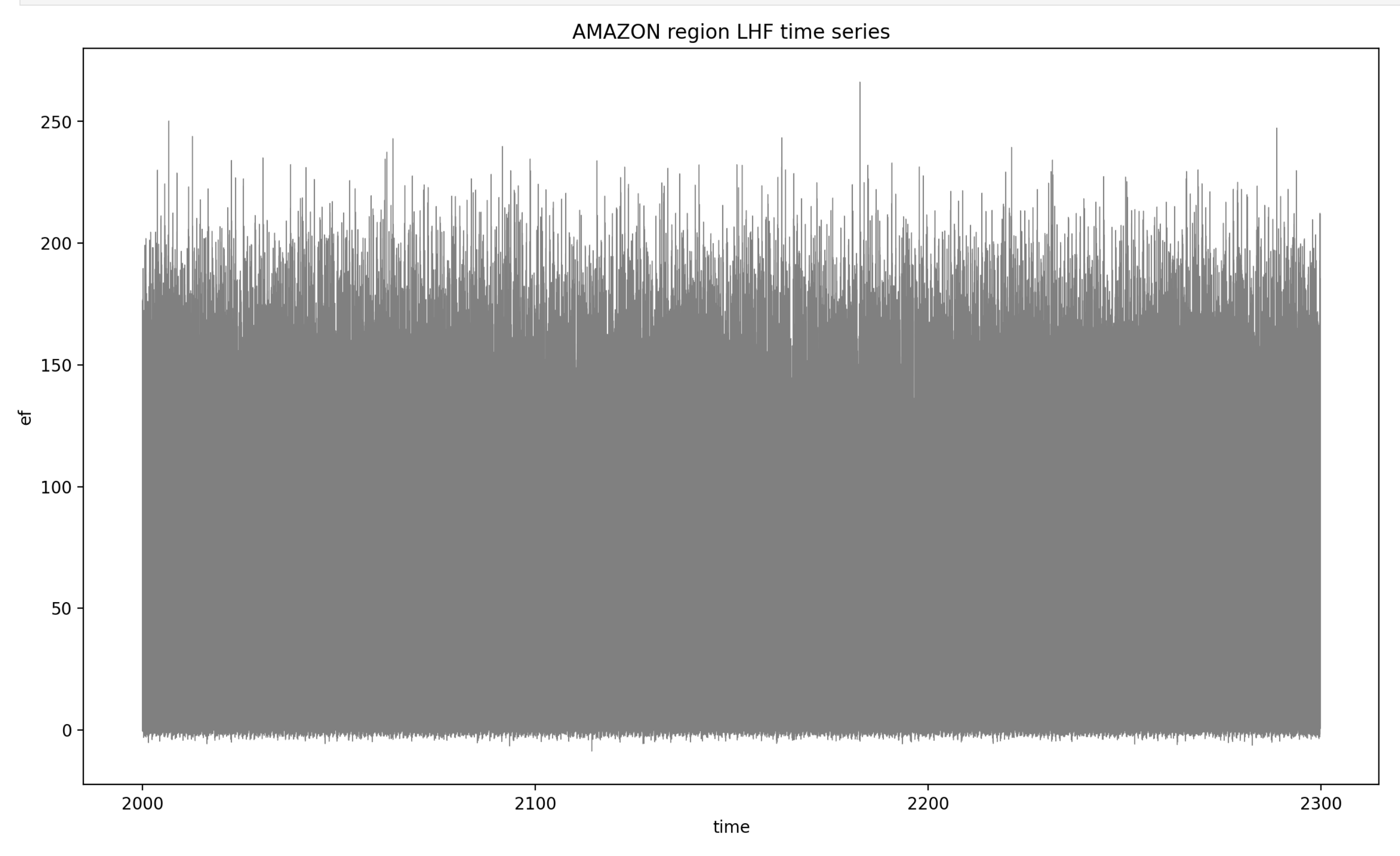
In [29]:

```
#plot the means
fig=plt.figure(figsize=(14,8))
AMAZON_region_mean.shf.plot(linewidth=0.5, color='grey')
plt.title("AMAZON region SHF time series")
#show plot
plt.show()
fig.savefig("SHF_run_AMAZON_"+runn+".png")
```



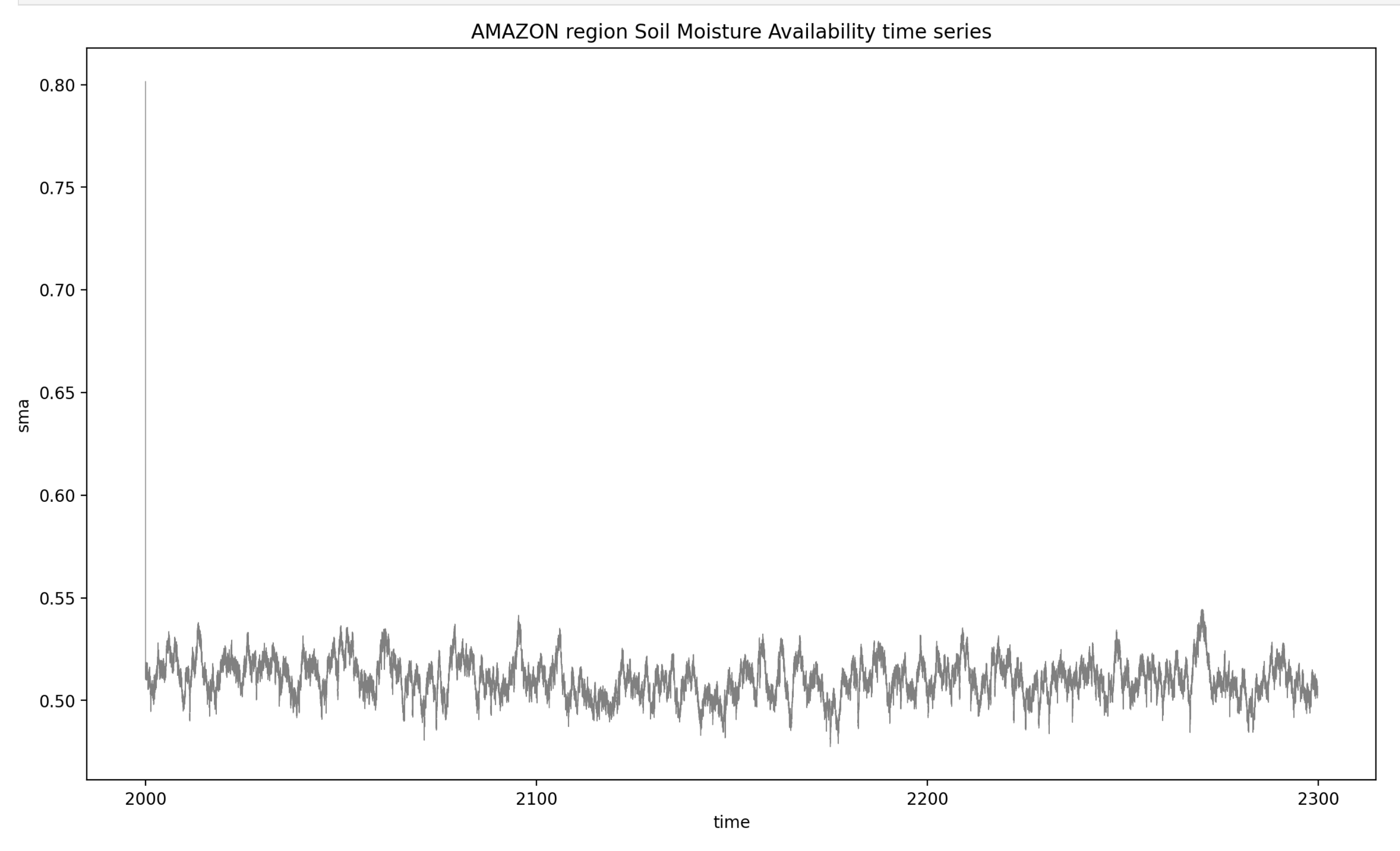
In [30]:

```
#plot the means
fig=plt.figure(figsize=(14,8))
lhf=2.5E6*AMAZON_region_mean.ef
lhf.plot(linewidth=0.5, color='grey')
plt.title("AMAZON region LHF time series")
#show plot
plt.show()
fig.savefig("LHF_run_AMAZON_"+runn+".png")
```



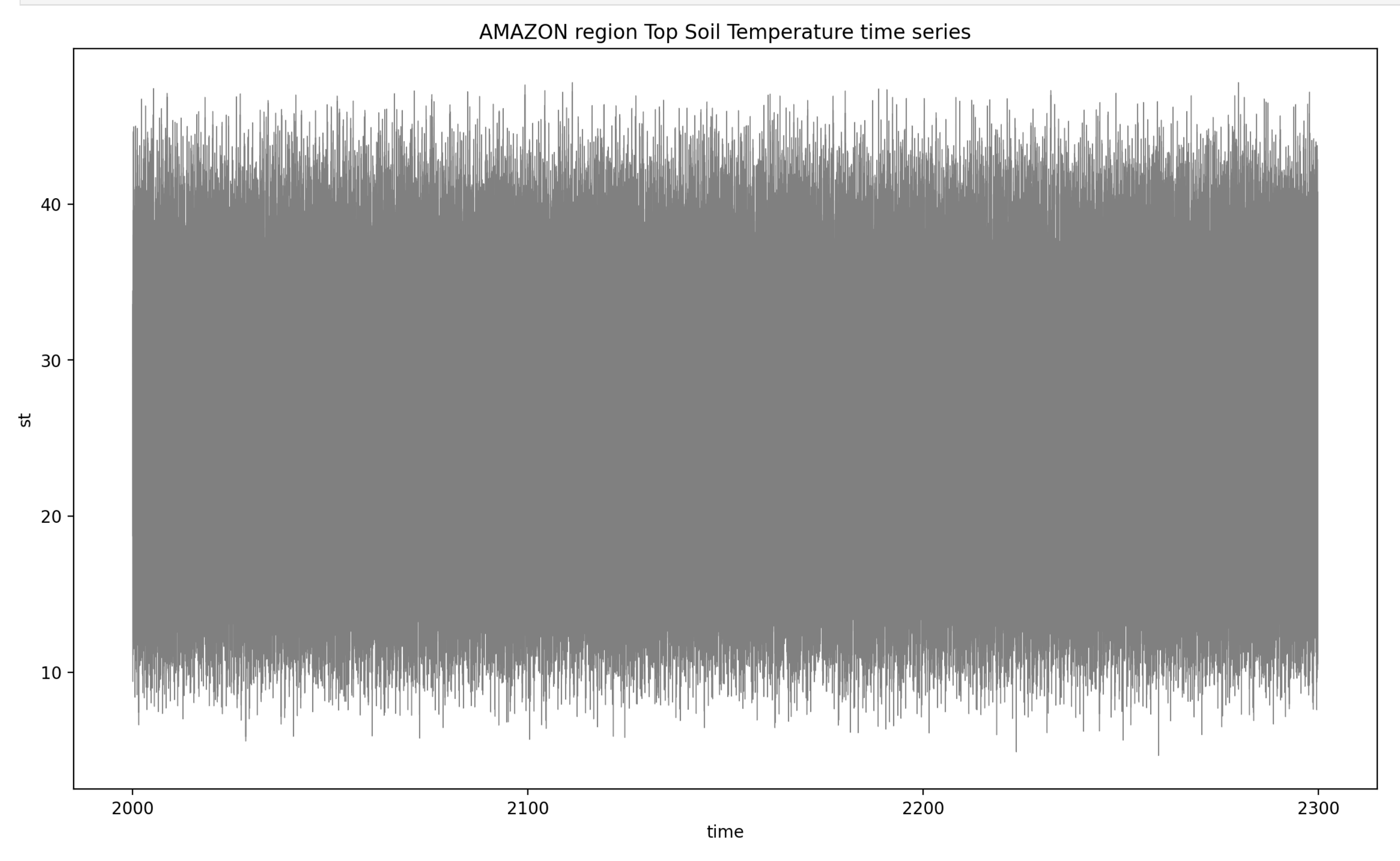
In [31]:

```
fig=plt.figure(figsize=(14,8))
AMAZON_region_mean.sma.plot(linewidth=0.5, color='grey')
plt.title("AMAZON region Soil Moisture Availability time series")
#show plot
plt.show()
fig.savefig("SMA_run_AMAZON_"+runn+".png")
```



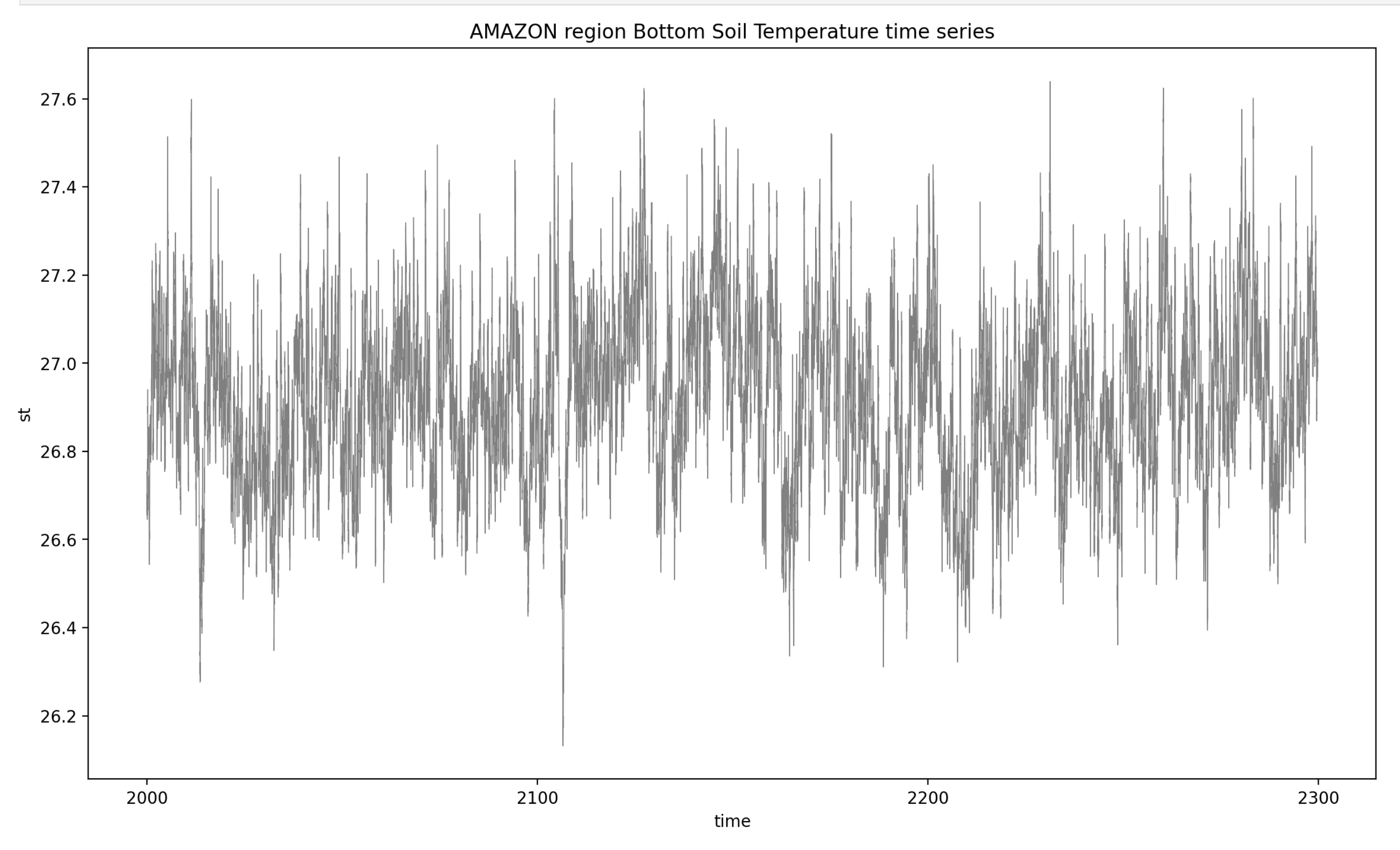
In [32]:

```
fig=plt.figure(figsize=(14,8))
st1=AMAZON_region_mean.st[:,0]
st1.plot(linewidth=0.5, color='grey')
plt.title("AMAZON region Top Soil Temperature time series")
#show plot
plt.show()
fig.savefig("ST1_run_AMAZON_"+runn+".png")
```



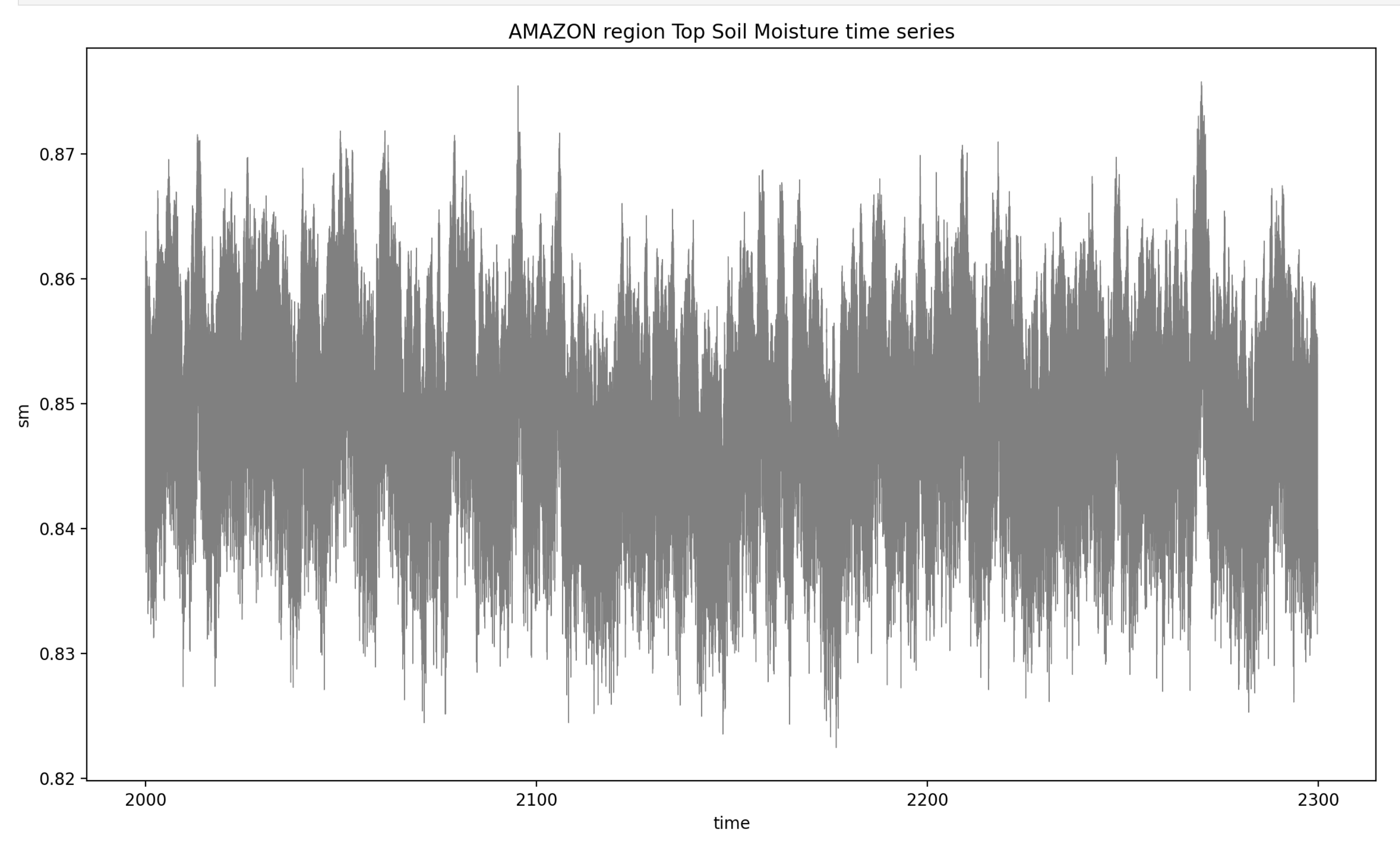
In [33]:

```
fig=plt.figure(figsize=(14,8))
st2=AMAZON_region_mean.st[:,1]
st2.plot(linewidth=0.5, color='grey')
plt.title("AMAZON region Bottom Soil Temperature time series")
#show plot
plt.show()
fig.savefig("ST2_run_AMAZON_"+runn+".png")
```



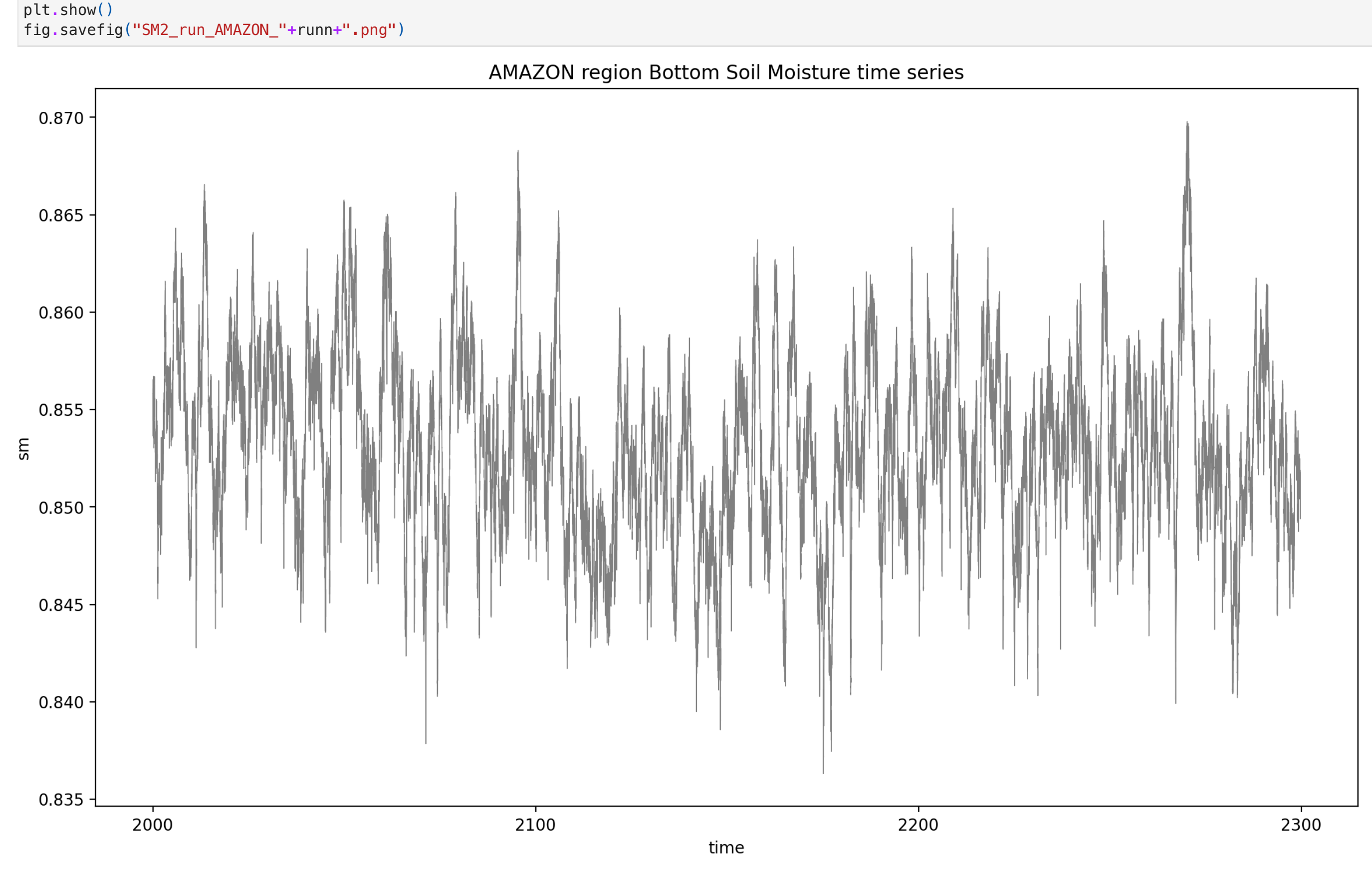
In [34]:

```
fig=plt.figure(figsize=(14,8))
sm1=AMAZON_region_mean.sm[:,0]
sm1.plot(linewidth=0.5, color='grey')
plt.title("AMAZON region Top Soil Moisture time series")
#show plot
plt.show()
fig.savefig("SM1_run_AMAZON_"+runn+".png")
```



In [35]:

```
fig=plt.figure(figsize=(14,8))
sm2=AMAZON_region_mean.sm[:,1]
sm2.plot(linewidth=0.5, color='grey')
plt.title("AMAZON region Bottom Soil Moisture time series")
#show plot
plt.show()
fig.savefig("SM2_run_AMAZON_"+runn+".png")
```



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